Math 208 Final, Winter 2024 — Problem 1

(a) $A\begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix} = \begin{bmatrix}5 & 7\\ 3 & 4\end{bmatrix}$ and $\begin{bmatrix}1 & 2\\ 2 & 5\end{bmatrix}^{-1} = \begin{bmatrix}5 & -2\\ -2 & 1\end{bmatrix}$:

$$\boxed{A = \begin{bmatrix}11 & -3\\ 7 & -2\end{bmatrix}} \quad \text{(check: } A(1,2) = (5,3),\ A(2,5) = (7,4)\text{)}.$$

(b) **False** (the zero map is not onto). **True** (a map $\mathbb{R}^4 \to \mathbb{R}^3$ has kernel of dimension ≥ 1). **False** (rows spanning $\mathbb{R}^m$ means rank m, but onto $\mathbb{R}^n$ needs rank n; e.g. $A = \begin{bmatrix}1\\ 0\end{bmatrix}$).

Unofficial worked solution — written by UWMathFinder, not by the instructor. Spot an error? Report it on the site.

Math 208 Final, Winter 2024 — Problem 2

Let $M = [\mathbf{v}_1\ \mathbf{v}_2\ \mathbf{v}_3]$; $\det M = (a_2 - 1)(a_1 - 3)$. If $\det M \neq 0$ the vectors span $\mathbb{R}^3$, so $\mathbf{b}$ is in the span.

If $a_1 = 3$: $\mathbf{v}_1 = \mathbf{v}_2 = (3, 1, 0)$, so every vector in the span has first two coordinates a multiple of $(3, 1)$; $\mathbf{b}$ has $(0, 2)$, so $\mathbf{b}$ is **not** in the span (for any a_2).

If $a_1 \neq 3$ but $a_2 = 1$: $(a_1, 1, 0)$ and $(3, 1, 0)$ span the xy-plane, which contains $\mathbf{b}$.

Answer: $\mathbf{b} \notin$ span exactly when $\boxed{a_1 = 3 \text{ } (a_2 \text{ arbitrary})}$.

Math 208 Final, Winter 2024 — Problem 3

(1) $A \sim \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, so $x_1 = x_3$, $x_2 = x_4$: $\ker T = \operatorname{span}\{(1,0,1,0),\ (0,1,0,1)\}$.

(2) Use the basis as columns: $S : \mathbb{R}^2 \to \mathbb{R}^4$,

$$S(\mathbf{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{x}.$$

Its columns are independent (so S is one-to-one) and span $\ker T$ (so $\operatorname{range}(S) = \ker T$).

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Math 208 Final, Winter 2024 — Problem 4

(1) For example $\begin{bmatrix}1&1&1\\1&1&1\\1&1&1\end{bmatrix}$: equal rows, so $\det A = 0$.

(2) The three given vectors span a 2-dimensional space ($(1,0,2,1) - (1,0,1,0) = (0,0,1,1)$ and $(1,0,3,2) - (1,0,1,0) = 2(0,0,1,1)$), so take them as the rows:

$$A = \begin{bmatrix}1&0&1&0\\1&0&2&1\\1&0&3&2\end{bmatrix}, \quad \text{rank } 2,\ \text{nullity } 4-2=2.$$

(3) **Impossible**: for a square matrix, one-to-one $\iff$ $\ker = \{\mathbf{0}\}$ $\iff$ rank 3 $\iff$ onto.

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Math 208 Final, Winter 2024 — Problem 5

(1) The polynomial has degree 4: $\boxed{n = 4}$, eigenvalues $-1, 0, 1, 2$.

(2) **No** (0 is an eigenvalue, so A is singular).

(3) $A + 2I$ has eigenvalues $1, 2, 3, 4$, none zero, so it is invertible: $\boxed{\operatorname{rank}(A + 2I) = 4}$.

(4) $A - I$ has eigenvalues $-2, -1, 0, 1$. Since $\lambda = 1$ is a simple eigenvalue, $\dim E_1 = 1$, so $\operatorname{nullity}(A - I) = 1$ and $\boxed{\operatorname{rank}(A - I) = 3}$.

(5) **Yes**: 4 distinct eigenvalues give 4 independent eigenvectors.

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Math 208 Final, Winter 2024 — Problem 6

$\operatorname{rank}(A^3) < 3 \iff \det(A^3) = (\det A)^3 = 0 \iff \det A = 0.$

$$\det A = a(a^2 - 1) - 3(a - 0) = a^3 - 4a = a(a - 2)(a + 2).$$

$$\boxed{a = 0,\ 2,\ -2}.$$

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Math 208 Final, Winter 2024 — Problem 7

(1) $\det(A-\lambda I) = (2-\lambda)\det\begin{bmatrix}-\lambda & 0\\ 3 & -\lambda\end{bmatrix} = (2-\lambda)\lambda^2$: $\boxed{\lambda_1 = 2,\ \lambda_2 = 0}$.

(2) $A - 2I = \begin{bmatrix}0 & 0 & 0\\ 0 & -2 & 0\\ 0 & 3 & -2\end{bmatrix}$ gives $x_2 = x_3 = 0$: basis $\{(1,0,0)\}$.

(3) $A\mathbf{x} = \mathbf{0}$: $2x_1 = 0$, $3x_2 = 0$: basis $\{(0,0,1)\}$.

(4) **No**: only 2 independent eigenvectors in $\mathbb{R}^3$ ($\lambda = 0$ has algebraic multiplicity 2 but geometric multiplicity 1).

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Math 208 Final, Winter 2024 — Problem 8

With $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ the columns of A, the columns of B are $2\mathbf{c}_1$, $\mathbf{c}_3$, $\mathbf{c}_2 + \mathbf{c}_1$:

$$\det B = 2\det[\mathbf{c}_1, \mathbf{c}_3, \mathbf{c}_2 + \mathbf{c}_1] = 2\det[\mathbf{c}_1, \mathbf{c}_3, \mathbf{c}_2] = -2\det A = -2.$$

($\mathbf{c}_1$ can be dropped from the third column since it repeats column 1; swapping two columns flips the sign.) Then $\det(2B) = 2^3 \det B = \boxed{-16}$.

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Math 208 Final, Winter 2024 — Problem 9

$\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_5$ are dependent exactly when some nonzero $\mathbf{x} \in \text{null}(A)$ has $x_3 = x_4 = 0$. A general null vector is $s(1,0,2,0,3) + t(0,4,0,5,0) + r(2,4,4,5,6)$; $x_3 = 2s + 4r = 0$ and $x_4 = 5t + 5r = 0$ give $s = -2r$, $t = -r$, and then every entry is 0 (indeed $(2,4,4,5,6) = 2(1,0,2,0,3) + (0,4,0,5,0)$). So the only such null vector is $\mathbf{0}$, and $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_5$ **are linearly independent**.